

Estimate, Measure, Compare

Our agreement for the year, and a first look at measurement

Physics · 1A · 9 September 2026

Our agreement for the year

(slide 1–2)

This is the first physics lesson we do together. Before we start with physics, I want to make clear how we will work this year.

What I expect from you

(slide 3)

Three things, no more.

1. Bring your notebook, your calculator and the week's worksheets to every lesson.
2. Lab reports and group work are marked as a group; tests and oral questioning are individual.
3. I expect respect: for me, behaviour counts more than marks.

How the year is organised

(slide 4)

The year is divided into three terms, and each has its own thread.

Term	Period	Topic
1	September – 2 December	measurement and experimental skills
2	December – 17 March	vectors and forces
3	March – 15 June	equilibrium of rigid bodies

The dates of Term 2 and Term 3 are still provisional: the final timetable comes out in October.

How you will be assessed

(slide 5)

Assessment	Weight
Midterm test (written)	1
End-of-term test (written)	1
Oral tests	0.5
Lab reports and group work	0.2
End-of-unit quizzes	feedback only, not graded

End-of-unit quizzes are not graded, but they are not free: each one gives you feedback, and that feedback deserves your attention.

Our signal

(slide 6)

A signal to ask for silence. I raise my hand and stay silent. You do the same within five seconds. No talking over, no exceptions. We use it for the first time very soon, in the activity that follows, when we will really need it.

Estimate. Measure. Compare.

(slide 7)

Three words for today's activity. The instructions are on the worksheet *Estimate, then measure*, not on the slide.

Step 1: before you touch any instrument, estimate a quantity by eye. **Step 2:** measure it for real. At the end we fill in the class value together. You can measure the same quantity more than once, on different objects and with different instruments; use as many rows as you like. The worksheet has a column for the estimate, one for the measurement, one for the uncertainty, which you write in your own words, and one for the class value. You choose the unit.

What you could measure: the mass of a pen, the thickness of a pen, the length of a chair, the lap time of the animation on the board, the fall time of a pen dropped from about one metre, the circumference of a bottle or a cup (this one is tricky to guess by eye), or anything else you are curious about, as long as it is a length, a mass or a time.

Three masses to compare

(slide 8)

Before estimating a mass it helps to calibrate your sense of it. Here are three reference points.

Object	Mass	Value
2-euro coin	8.5 g	official specification, fixed
iPhone 13 mini	141 g	manufacturer specification, fixed
an apple	150 g–200 g	varies: no single true value

Sources: European Central Bank coin specification; Apple technical specifications.

Before moving on. Of the three masses in the table, only one has no true value. Which one, and why?

The apple. The coin and the phone leave the factory with a mass set by a specification; every apple is different from the next, so 150 g–200 g is a range, not a single number. It is already a first meeting with uncertainty: we will come back to it.

Counting without a stopwatch

(slide 9)

To estimate a time without a stopwatch we count like this: *one-Mississippi, two-Mississippi, three-Mississippi...* Saying the full word *Mississippi* takes about one second, so each *number-Mississippi* is roughly one second. It is not perfectly accurate, but it is far better than just counting numbers, which people tend to rush.

Before moving on. How long does one full lap of the animation on the board last? Estimate it by counting Mississippis, write the estimate on the worksheet, and only then time it.